

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B.Tech AI &DS	Date:	29-07-2026

Code of Conduct

I _____ N.Kishan kumar _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Q1. Morphological Analysis Pipeline

Input Words:

- Connected
- Connecting
- Connection

Step 1: Decompose into Root and Suffix

Word	Root	Suffix
Connected	Connect	-ed
Connecting	Connect	-ing
Connection	Connect	-ion

Step 2: Classification

- **Connected** → Inflectional (Past tense)
- **Connecting** → Inflectional (Present participle)
- **Connection** → Derivational (Forms a noun)

Step 3: Normalization

All words are converted to the common base form:

Connect

Parsed Structure

- Connected = Connect + ed
- Connecting = Connect + ing
- Connection = Connect + ion

Final Output

Normalized Root = Connect

Q2. Morphological Parsing Module

Input Words

- Unhappy
- Happiness
- Happily

Step 1: Morphological Breakdown

Word	Prefix	Base Word	Suffix
Unhappy	un-	happy	—
Happiness	—	happy	-ness
Happily	—	happy	-ly

Step 2: Classification

- Unhappy → Derivational
- Happiness → Derivational
- Happily → Derivational

Semantic Effect

- **un-** changes meaning to opposite.
- **-ness** changes adjective into noun.
- **-ly** changes adjective into adverb.

Final Root

Happy

Q3. Stemming-Based Preprocessing

Input Words

- Played
- Player
- Playing

Step 1: Remove Affixes

Word	Stem	Affix
Played	Play	-ed
Player	Play	-er
Playing	Play	-ing

Step 2: Classification

- Played → Inflectional
- Player → Derivational
- Playing → Inflectional

Final Stem

Play

Result

All words are indexed as:

Play

Q4. Finite-State Morphological Parsing

Input Words

- Writes
- Writing
- Written

Morphological Breakdown

Word	Root	Suffix	Type
Writes	Write	-s	Regular Inflection
Writing	Write	-ing	Regular Inflection
Written	Write	-en	Irregular Inflection

State Transitions

Start State → Root (**Write**)

- Write → +s → Writes
- Write → +ing → Writing
- Write → +en → Written

Normalized Root

Write

Result

All forms map to:

Write

Q5. Porter Stemmer-Based Processing

Input Words

- Relational
- Relation
- Relate

Step-by-Step Stemming

Word	Remove Suffix	Intermediate Form	Final Stem
Relational	-ational	Relate	Relat
Relation	-ion	Relat	Relat
Relate	-e	Relat	Relat

Explanation

- Remove derivational suffixes.
- Apply Porter Stemmer rules.
- Convert all words to one common stem.

Final Stem

Relat

Result

All words are indexed as:

Relat

Conclusion

Morphological analysis and stemming help convert different forms of a word into a common root or stem. This improves document indexing, searching, and information retrieval by treating related words as the same term.

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____